STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-COMPARE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1429-1452 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 1454-1505, the complete author proof (50 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_algebraic-stacks-sheaves.json (pb=2261, stmt 1429-1452, proof 1454-1505).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 02ZV: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1505–1505. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04SU: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1505–1505. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00XT: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1505–1505. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00XS: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1505–1505. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04D2: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1505–1505. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00XR: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1505–1505. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. Assume $\mathcal{X}$ is representable by an algebraic space F. Then there exists a continuous and cocontinuous functor $F_{étale} \to \mathcal{X}_{étale}$ which induces a morphism of ringed sites*

$$\pi_F : (\mathcal{X}_{étale}, \mathcal{O}_{\mathcal{X}}) \longrightarrow (F_{étale}, \mathcal{O}_F)$$

and a morphism of ringed topoi

$$i_F : (Sh(F_{étale}), \mathcal{O}_F) \longrightarrow (Sh(\mathcal{X}_{étale}), \mathcal{O}_{\mathcal{X}})$$

such that $\pi_F \circ i_F = id$. Moreover $\pi_{F,*} = i_F^{-1}$.

Proof. Choose an equivalence $j : \mathcal{S}_F \to \mathcal{X}$, see Algebraic Stacks, Sections 04SU and 02ZV. An object of $F_{étale}$ is a scheme U together with an étale morphism $\varphi : U \to F$. Then φ is an object of $\mathcal{S}_F$ over U. Hence $j(\varphi)$ is an object of $\mathcal{X}$ over U. In this way j induces a functor $u : F_{étale} \to \mathcal{X}$. It is clear that u is continuous and cocontinuous for the étale topology on $\mathcal{X}$. Since j is an equivalence, the functor u is fully faithful. Also, fibre products and equalizers exist in $F_{étale}$ and u commutes with them because these are computed on the level of underlying schemes in $F_{étale}$. Thus Sites, Lemmas 00XR, 00XS, and 00XT apply. In particular u defines a morphism of topoi $i_F : Sh(F_{étale}) \to Sh(\mathcal{X}_{étale})$ and there exists a left adjoint $i_{F,!}$ of i_F^{-1} which commutes with fibre products and equalizers.

We claim that $i_{F,!}$ is exact. If this is true, then we can define π_F by the rules $\pi_F^{-1} = i_{F,!}$ and $\pi_{F,*} = i_F^{-1}$ and everything is clear. To prove the claim, note that we already know that $i_{F,!}$ is right exact and preserves fibre products. Hence it suffices to show that $i_{F,!}* = *$ where $*$ indicates the final object in the category of sheaves of sets. Let U be a scheme and let $\varphi : U \to F$ be surjective and étale. Set $R = U \times_F U$. Then

$$h_R \rightrightarrows h_U \longrightarrow *$$

is a coequalizer diagram in $Sh(F_{étale})$. Using the right exactness of $i_{F,!}$, using $i_{F,!} = (u_p)^\#$, and using Sites, Lemma 04D2 we see that

$$h_{u(R)} \rightrightarrows h_{u(U)} \longrightarrow i_{F,!}*$$

is a coequalizer diagram in $Sh(\mathcal{X}_{étale})$. Using that j is an equivalence and that $F = U/R$ it follows that the coequalizer in $Sh(\mathcal{X}_{étale})$ of the two maps $h_{u(R)} \to h_{u(U)}$ is $*$. We omit the proof that these morphisms are compatible with structure sheaves. □